

WireGuard Setup Guide

Soccer Vision project | subnet 10.10.10.0/27 | hub-and-spoke topology

This tunnel links the two project servers on the 10.10.10.0/27 subnet. The EC2 instance is the hub and the only node with a public endpoint; the **soccer-vision-vm** attaches to it as a peer. Run every command as **root**. Replace each `<...>` placeholder with the real value for the node you are configuring.

1 Common setup — every node

Enable IPv4 forwarding (required on the EC2 hub so it can route traffic between peers; harmless on the leaf nodes), install the WireGuard tooling, then generate this node's keypair.

```
echo "net.ipv4.ip_forward = 1" > /etc/sysctl.d/99-wireguard.conf
sysctl --system
apt install wireguard-tools
wg genkey | tee /etc/wireguard/$HOSTNAME.private.key | wg pubkey > /etc/wireguard/$HOSTNAME.pub
chmod 600 /etc/wireguard/$HOSTNAME.private.key /etc/wireguard/$HOSTNAME.public.key
```

2 EC2 instance (hub) — /etc/wireguard/wg0.conf

The hub owns the public ListenPort. SaveConfig = false keeps this file authoritative — the peer is edited in by hand, not persisted by the runtime. The single [Peer] block is the soccer-vision-vm.

```
[Interface]
Address = 10.10.10.1/27
ListenPort = 51820
PrivateKey = <EC2 private key>
SaveConfig = false

[Peer]
PublicKey = <soccer-vision-vm pub key>
AllowedIPs = 10.10.10.2/32
```

3 soccer-vision-vm (peer) — /etc/wireguard/wg0.conf

The VM dials the hub by its public IP. AllowedIPs = 10.10.10.0/27 routes the whole tunnel subnet through the hub; PersistentKeepalive holds the path open through NAT.

```
[Interface]
Address = 10.10.10.2/32
PrivateKey = <soccer-vision-vm private key>

[Peer]
PublicKey = <EC2 public key>
Endpoint = <EC2 public IP>:51820
AllowedIPs = 10.10.10.0/27
PersistentKeepalive = 25
```

4 Enable and verify — both nodes

Bring the interface up now and on boot, then confirm the handshake and transfer counters.

```
systemctl enable wg-quick@wg0 --now
wg show
```

5 Address allocation — 10.10.10.0/27

Node	Role	WireGuard address
EC2 instance	Hub / public endpoint	10.10.10.1
soccer-vision-vm	Peer	10.10.10.2